

Lecture Notes

EC226 Term 1 2025/26

Multicollinearity

Readings:

- Stock and Watson (2003): Chapter 6
- Dougherty (2016) Chapter 6
- Wooldridge (2013): Chapter 3

Contents

1	Perfect multicollinearity	3
2	Imperfect multicollinearity	4
2.1	Properties of Estimators	4
2.2	Detecting Multicollinearity	4
2.3	Solutions for (imperfect) multicollinearity	4

1 Perfect multicollinearity

Consider the model,

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_k x_{ki} + \varepsilon_i \quad i = 1, 2, \dots, n \quad (1)$$

perfect multicollinearity arises when one of the x variables, for example, $x_{j=1i}$ is an exact linear function of some other x variables in the model, for example $x_{1i} = 2 + 3x_{2i}$. Now equation (1) cannot be estimated by OLS as the variables x_{1i} and x_{2i} cannot be distinguished from each other as it is impossible to $\frac{\partial y_i}{\partial x_{1i}} = \beta_1$ (holding all else constant) as x_{2i} changes exactly as x_{1i} changes.

Perfect multicollinearity should never arise and often reflects a mistake on the part of the researcher.

The solution in our example above would be to drop either x_1 or x_2 .

Examples of perfect multicollinearity:

$$\ln(M_i) = \beta_0 + \beta_1 \ln(GDP_i) + \beta_2 Inf_i + \beta_3 r_i + \beta_4 rr_i + \varepsilon_i \quad (2)$$

where M = Real money, GDP = Real gross domestic product, Inf = Inflation rate, r = Nominal interest rate, rr = Real interest rate. However, as $rr_i = r_i - Inf_i$, we have perfect multicollinearity and we cannot estimate all the parameters of the model. The solution would be to include any two of the three variables Inf , r , rr , from equation (2) and estimate:

$$\ln(M_i) = \beta_0 + \beta_1 \ln(GDP_i) + \beta_2 Inf_i + \beta_3 r_i + \varepsilon_i$$

Alternatively, returning to Handout 3 (on Dummy variables), consider the problem with estimating:

$$\ln(w_i) = \beta_0 + \beta_1 School_i + \beta_2 F_i + \beta_3 NF_i + \varepsilon_i \quad (3)$$

where, w = wages, $School$ = Number of years of schooling, $F = 1$ if female, 0 if non-female and $NF = 1$ if non-female, 0 if female. But as $NF = 1 - F$, we have perfect multicollinearity consider the actual data we might have for our series:

$\ln(w)$	Intercept	School	Female	Non-female
2.84	1	13	0	1
2.45	1	11	0	1
4.12	1	16	0	1
1.35	1	11	0	1
1.86	1	13	0	1
2.24	1	11	1	0
2.56	1	11	1	0
2.68	1	13	1	0
2.50	1	16	1	0

And we can see immediately that the NF variable can be constructed as the variable for the Intercept minus the F variable.

2 Imperfect multicollinearity

As perfect multicollinearity should only occur in error, when we talk about multicollinearity we are generally referring to severe imperfect multicollinearity. Imperfect multicollinearity occurs when there is a very high correlation between two or more explanatory variables (although this correlation is not unity as this would be perfect multicollinearity).

2.1 Properties of Estimators

1. The OLS estimates remain unbiased.
2. Since two (or more variables) are strongly related it becomes increasingly difficult to identify their separate effects such that the standard error on the coefficient estimates (our measure of uncertainty on the coefficients) will become increasingly larger and the t-ratios will be smaller and we are therefore much more likely to accept any null hypothesis (whether it is true or not).

Remember the idea of partitioned regression we can write the variance of the coefficient b_1 in the multivariate regression as:

$$V(b_1) = \frac{\sigma^2}{\sum_{i=1}^n \tilde{x}_{1i}^2} \quad (4)$$

where $\sum_{i=1}^n \tilde{x}_{1i}^2$ is that variation in x_1 not accounted for by the other variables $x_{2i}, \dots, x_{ki}$.

If there is at least one other variable which looks like, x_1 then these other variables capture most of the variation in x_1 and hence the residuals from a regression of x_1 on these variable are very small, implying $\sum_{i=1}^n \tilde{x}_{1i}^2 \Rightarrow 0$. Given equation (4) we therefore know $V(b_1) \Rightarrow \infty$.

Multicollinearity may also be reflected by the parameter estimates of any model being very sensitive to small changes in the specification of the model. The overall R^2 of an equation is largely unaffected by multicollinearity.

2.2 Detecting Multicollinearity

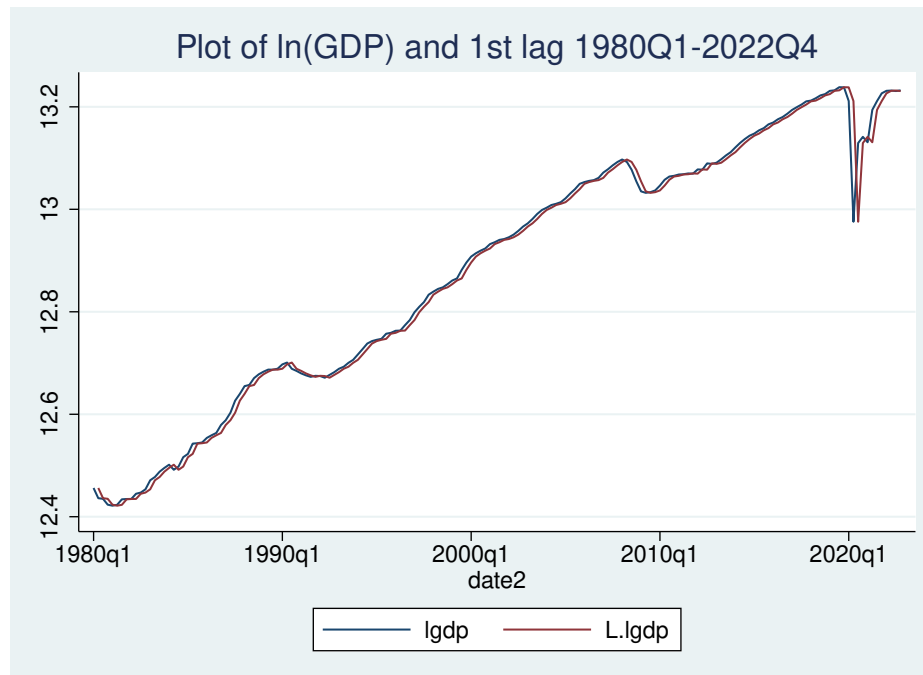
The variance inflation factor (VIF) is a statistical measure used in regression analysis to detect multicollinearity. VIF is calculated for each explanatory variable in turn as $1/(1 - R^2)$, where R^2 is the value from a regression of that explanatory variable against all the other explanatory variables in the model.

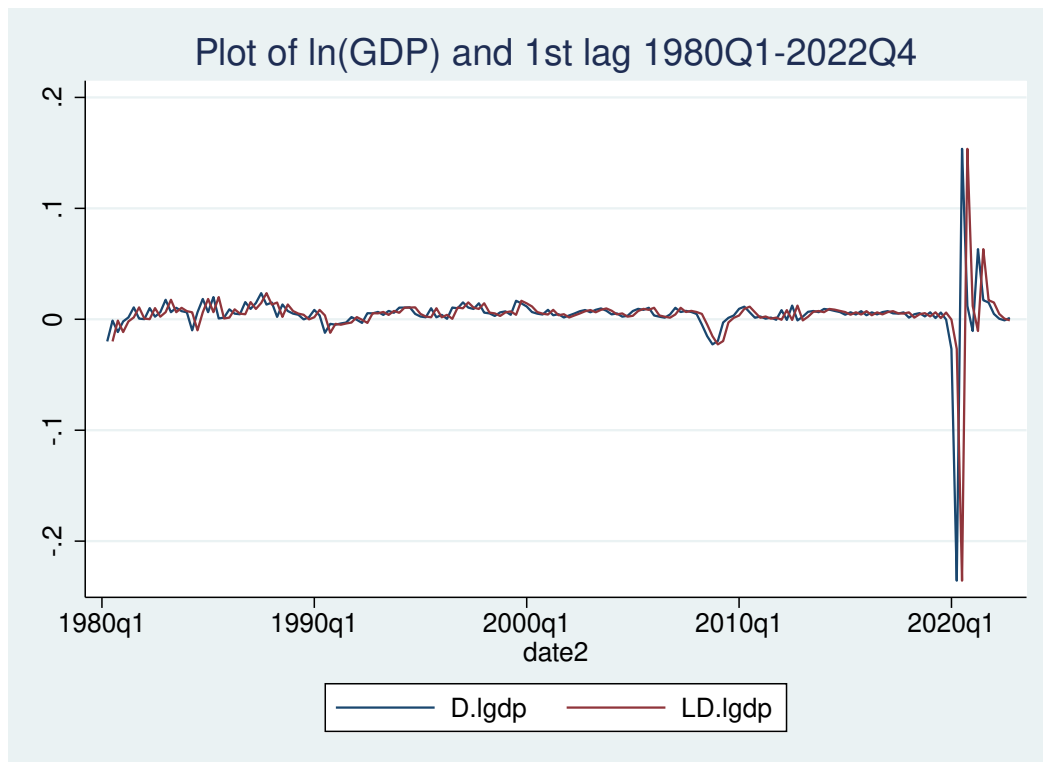
- A VIF of 1 indicates no multicollinearity.
- Higher VIF values indicate greater multicollinearity.
- A common rule of thumb is that VIF values above 5 or 10 suggest a problematic level of multicollinearity.

2.3 Solutions for (imperfect) multicollinearity

1. Do nothing this is the simplest and most appropriate solution if the variables in which we are interested are significant and the coefficient estimates not excessively sensitive to small changes in the specification of the model.

2. Drop the variable which we believe to be highly collinear with one (or more) other variables. If the variables are very collinear then the choice as to which to drop is essentially arbitrary (as the fit of the model having dropped each in turn will be approximately similar) the and the choice ought to be based on the theoretical model underlying your empirical model.
3. Increase the sample size and hope this fixes up the problem – although this might be fanciful as this is potentially very costly in terms of time and effort.
4. Transform the collinear variables, by:
 - (a) Form a combination of the collinear variables, using, for example, principle component analysis (PCA).
 - (b) In time series data, where the unit of measurement is time, then as most time series variables trend upwards over time (e.g. GDP, Consumption, Exports, Imports, etc) one solution might be to take first differences and use the change in GDP instead, for example, using quarterly data from 1980:1-2022:4, the correlation between GDP_i and GDP_{i-1} is 0.996, but that between ΔGDP_i and ΔGDP_{i-1} is only -0.257. This can also be seen in the two Figures below.





References

Dougherty, C. (2016). *Introduction to Econometrics*. OUP Catalogue. Oxford University Press.

Stock, J. and Watson, M. W. (2003). *Introduction to Econometrics*. Prentice Hall, New York.

Wooldridge, J. M. (2013). *Introduction to Econometrics*. Cengage.